Learnings & Reflections Document

1. Team Reflection

- *How did the team work together?*
- We divided the project early and clearly ,we assigned specific trials and hardware sections between us, with this, we could work parallely without interference. One person had the responsibility of the goblet and IR detection, the dragon trial and servo mechanics, and another handled the sequence game and the gillyweed trial. This split made starting much easier and removed a lot of the initial confusion.
- The code was written individually per trial and then merged into a unified single code sta at the end. This worked well for development speed but it created a real challenge during integration everything together

- *What worked well?*

- Early task division was of the biggest help, so we could take ownership over our work, and held responsible for it

- We were able to adapt quickly when hardware failed, when the power supply module got fried with no visual indication, we tested it with a multimeter and quickly were able to replace it and start our physical setup

- The physical and code work happened simultaneously at the same time, while the code was being written and tested on breadboards, the cardboard and paper mache build was being constructed. This saved us a lot of time in the final week and assembly.

- We were able to conduct the entire flow of the game with no debugging or changing of setup, there were no fails, the system worked for 20+ runs in the final jury

- *What could be improved?*

- More combined working sessions would have help us get caught integration problems earlier. Because each person worked mostly independently, the combining stage at the end was more stressful than it needed to be.

- Documentation was neglected for a while, earlier more documentation would have been better

- Not enough time to run a lot of test trials with different users

2. Technical Reflection

- *Key technical challenges*
- *What did you learn (hardware/software)? Mistakes & debugging experiences*

- One of our power supplies blew during testing. There was no visible sign, no burn marks. The only way we caught it was by measuring voltage output with a multimeter and finding 12V where we expected 5V. This caused two neopixel strips to fail. Replacing the components took about a week which was our tightest constraint
- During our first soldering attempt, we connected to the DO side of the neopixel strip instead of the DI side. The strips didn't respond at all, and it took longer than it should have to diagnose because the code looked completely correct. And another soldering issue we faced during jury trial was that our negative and positive soldering were touching each other which caused the entire neopixel strip to fail
- The servo stick that was supposed to pull the dragon's mouth open was too short and the mouth barely moved. We extended it using a foam board stick taped on, which increased the lever arm and gave us the full dramatic opening we wanted
- Initially, the BLE advertising only started when Trial 2 began. By then, players had no time to scan, connect, and be ready before the sequence started. We moved advertising to the very start of the program during the idle state, which meant the player could connect their phone and be fully ready before pressing the start button.
- Always verify voltage with a meter before connecting components, especially when changing power supplies.
- Neopixel strips are far more pin efficient than individual LEDs for multi LED setups because it just needs one data pin controls multiple leds
- BLE on the ESP32 needs to advertise early
- Start working on combining code early on

3. Design Reflection

- *Did the experience match your intention?*

The main goal for us was to make something that felt like a real game with a narrative thread running through it, and not just multiple sensors. The goblet entry, the dragon egg reveal, the sequence trial, the gillyweed harvest, and the trophy, each one connects to the next in a way that tells the story of the Triwizard Tournament rather than just presenting different games.

- The servo dragon mouth opening was the moment we were most proud of. It added a moving element and it reveals the key to the next trial.
- All the elements worked in conveying the story through coding
- *What would you improve?*
- The trophy reveal was supposed to be a stepper motor raising the trophy from behind the game table but we had to abandon this because the trophy was too heavy. If we had more time, we would have built a lighter trophy prop and used a stronger motor
- More GPIO pins would have let us add more physical buttons and sensors rather than using some interactions through the MIT app. The Bluetooth input works but it had to be connected each time the code was ran and sometimes it disconnected, so not very reliable.
- DFPlayer Mini with proper MP3 audio would have elevated the atmosphere significantly and improved storytelling
- *User feedback insights*
- The goblet and dragon flow was the most apperiacted part in our entire game, with the servo attracting most players
- The third game required a lot of instructions despite the written visual cues
- Players liked the different types of games and different testing
- The amount of interactions was most appreiciated

4. Personal Reflection

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- *What they learned*
- ESP32 and power supply module were powerful enough for a full multi trial game system, without the need for any external battery, we were worried if our 110 neopixel setup would be supported by the power supply but it worked.
- The physical build and the code need to be developed in parallel and tested together as early as possible
- *What they struggled with*
- The power supply failure was the most stressful because of the time lost waiting for replacement components and not knowing initially what had failed
- Combining different codes having while true loops
- Keeping the physical build durable enough for repeated demo plays
- *What they are proud of*
- The entire setup of the tournament worked in with go , without any elements failing. Even though we had a lot of individual elements that needed to work right to continue,

they all fell into place and we were able to execute it flawlessly without any failures the 20+ times it was played.

- The storytelling was consistent throughout the entire game.especially the egg inside the dragon's mouth that would be required to start the next trial.
- Solving the BLE timing problem and effective use of gpio pins.